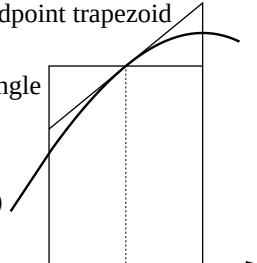


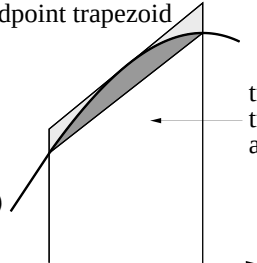
midpoint trapezoid

midpoint rectangle

$$y = f(x)$$


This diagram illustrates two numerical integration methods for a function $y = f(x)$. A solid black curve represents the function. A vertical line segment from the x-axis to the curve at its midpoint defines the height of a rectangle, labeled 'midpoint rectangle'. A trapezoid is formed by the function curve, the x-axis, and two vertical lines at the interval endpoints, labeled 'midpoint trapezoid'. A dashed vertical line connects the midpoint of the x-axis to the curve.

midpoint trapezoid

$$y = f(x)$$


This diagram shows the same function $y = f(x)$ and midpoint trapezoid approximation. The area between the function curve and the top edge of the trapezoid is shaded gray. An arrow points from the text 'trapezoid from trapezoidal approximation' to this shaded region.

trapezoid from
trapezoidal
approximation